10. Applying Performance Optimization Techniques (Indexing, Subforms, and Query Tuning) in Zoho Creator

Aim:

To implement performance optimization techniques, including field indexing, subform structuring, and Deluge query tuning, in the Insurance Claim Request & Processing Management application to improve execution speed, reduce response latency, and optimize database operations.

Overview:

As low-code applications grow and handle larger volumes of data, inefficient data structures and poorly written scripts can cause performance issues. Zoho Creator provides several techniques to optimize application performance.

1. Field Indexing:

Field indexing improves the speed of searching, filtering, and retrieving records.
In the insurance claim application, frequently searched fields such as Policy Number, Claimant Name, and Claim Status can be considered for indexing where supported.

Benefits:

- Faster record searches.
- Improved filtering performance.
- Reduced database lookup time.

2. Efficient Subform Usage

Subforms allow related information to be stored within a parent form.
In the insurance claim application, an Investigation Details subform could store investigation dates, surveyor details, findings, and investigation status for each claim.

B**enefits:**

- Organizes related information.
- Simplifies data entry.
- Helps manage investigation records efficiently.

3. Deluge Query Tuning

Deluge scripts can be optimized by retrieving only the required records, using precise criteria, and avoiding unnecessary database operations inside loops.

For example, instead of fetching every claim, retrieve only claims that match a particular status or date range.

Benefits:

- Reduces unnecessary database operations. i
- improves script execution time.
- Helps prevent execution timeouts.

Procedure

Step 1: Identify Fields for Indexing

1. Open the Insurance Claim Request & Processing Management application in Zoho Creator.
2. Open the Claim Request Form in the Form Builder.
3. Identify frequently searched fields, such as Policy Number and Claim Status.
4. Open the relevant field properties.
5. Enable indexing or uniqueness only if the option is available and appropriate. Do not make Policy Number unique unless the application's policy rules require it.

Step 2: Structure Subforms

1. Open the Claim Request Form in the Form Builder.
2. Consider adding an Investigation Details subform if multiple investigation entries must be recorded for each claim.
3. Add relevant fields, such as Investigation Date, Surveyor Name, Findings, and Investigation Status.
4. Include only necessary fields.
5. Configure appropriate validations and save the form.

Step 3: Optimize Deluge Scripts

1. Open the workflow or form event that retrieves claim records.
2. Review existing Deluge scripts for unnecessary database queries.
3. Use specific criteria to retrieve only relevant claims.
4. Avoid database fetches and inserts inside loops when batch processing is possible.
5. Save the script and test its execution.

Example Deluge query:

```
claims = Claim_Request_Form[Claim_Status == "Submitted"];
```

This retrieves claims whose status is Submitted, rather than retrieving all claims.

Step 4: Test and Monitor Performance

1. Open the application debugger and execution logs, where available.
2. Record script execution times before and after optimization.
3. Test report loading and form submission.
4. Check that optimized queries return the correct records.
5. Verify that existing workflows and validations continue to work.

Result

Performance optimization techniques were identified for the Insurance Claim Request & Processing Management application. Field indexing can improve searches, structured subforms can organize investigation details, and optimized Deluge queries can reduce unnecessary database operations.